2.3.2 Poisson's distribution.

Discrete random variable *X* has *Poisson's distribution* if it has infinite calculating set of possible values 0, 1, 2..., *m*, …, and probabilities corresponding them are defined by the formula:

$$P_m = \frac{\lambda^m \cdot \exp(-\lambda)}{m!}, \qquad m = 0,1,2,\ldots \qquad (23)$$

n = max(m) > 2λ

Examples of the casual phenomena, subordinated to Poisson's distribution law, are: sequence of radioactive disintegration of particles, sequence of refusals at work of complex computer system, a stream of applications on a telephone exchange and many other things.

Poisson's distribution law (23) depends on one parameter *λ* which simultaneously is both a mathematical expectation, and a dispersion of random variable *X* . Thus, for Poisson's distribution law the following basic numerical characteristics take place:

Mathematical expectation: $E(X) = \mu_X = \lambda,$

Dispersion: $D(X) = \lambda$ ***Standard deviation:*** $\sigma = \sqrt{\lambda}$

Example 2.6

With MAPLE construct *Poisson's* distribution for *n=15* and

λ=0.5;

λ=5.;

λ=15.;

Construct histogram.

Find mean or expected value $E(X) = \mu_X$

Find variance (dispersion) and standard deviation.